

ON THE CONTROL EFFICIENCY OF MULTI-AGENT LANGUAGE MODEL SYSTEMS

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ABSTRACT

How efficiently can an external controller steer a group of interacting language-model agents? We study this question in relational reasoning tasks where evidence is distributed across the population. Our feedback controller has two separate limits: it observes only a sample of the agents and can intervene in only a fixed number of their interactions. This setup lets us measure the distinct effects of sensing and actuation. We introduce finite-time measures for three aspects of control: how much an intervention changes the population, how efficiently action-dependent information becomes coherent population motion, and how efficiently the finite-time thermodynamic control budget becomes directed current. We derive a finite-state reference theory with a single measurable directional affinity h and a kinetic compliance γ , which separates which direction controlled revisions favor from how readily agents revise at all. In experiments with matched initial social and knowledge states, advocacy increases support for the controller’s target in every tested condition, with susceptibilities from 0.100 to 0.163. The transmitted control information depends strongly on the population state. More actuation produces a wider informative region, whereas stronger feedback gain is predicted to concentrate information near the controller’s decision boundary. Information–response efficiency increases with actuation across the full resource grid, from about 0.4% to 2.5% of the Pinsker upper bound. A calibrated finite-compliance model also reproduces the observed controlled current closely ($r = 0.999$). Finally, the single-affinity theory yields the exact finite-time identity $\Delta S_{\text{sys}} + hJ_c + I(n_k; Y_k) = \Sigma \geq 0$ and the bounded thermodynamic efficiency $\eta_{\text{th}} = hJ_c/[hJ_c + I(n_k; Y_k)]$. The resulting theory predicts diminishing returns from additional sensing whenever the acquired information does not produce a proportional increase in directed current. Our framework measures control efficiency while keeping it separate from whether the agents reach the correct answer.

1 INTRODUCTION

Large language models (LLMs) are increasingly deployed not only as individual problem solvers, but as components of *multi-agent systems* in which many autonomous agents communicate, exchange information, and collectively determine an outcome. Recent work has shown that populations of LLM agents can exhibit genuinely collective phenomena, including spontaneous convention formation, amplification of population-level biases, and higher-order coordination that cannot be understood from isolated agents alone (Ashery et al., 2025; Riedl, 2026). These observations raise a complementary question that becomes increasingly important as such systems are deployed: *how can an external agent control a collective of interacting LLMs?* In particular, we are interested not in unconstrained steering—where a controller can inspect and modify the entire system—but in the more realistic regime in which the controller itself has limited resources.

We study this problem through a *bounded feedback controller*. At each population round, the controller observes only a finite random sample of agents and must use this partial observation to decide whether to intervene. Its sensing capacity is therefore limited: the controller never observes the complete population state and its estimate is intrinsically noisy. Its actuation capacity is limited as well: even when it decides to act, it may replace the social input in only a bounded number of interactions. The resulting closed loop has the form

$$N_t \longrightarrow Y_t \longrightarrow U_t \longrightarrow N_{t+1}, \quad (1)$$

where N_t denotes the population state, Y_t the controller’s partial observation, and U_t its intervention. This formulation makes imperfect information a consequence of finite sensing resources rather than an artificial source of observation noise, and separates two distinct constraints on control: *how much of the system can be observed* and *how much of it can be acted upon*. Our central question is therefore:

How efficiently can a resource-constrained feedback controller steer an interacting population of LLM agents when both information and actuation are limited?

Controlling an LLM population, however, should involve more than controlling a population of stochastic finite-state machines. The distinctive feature of language-model agents is precisely that their interactions can carry *semantic information*: agents can possess different facts, communicate those facts through language, accumulate evidence, and reason over combinations of information. We therefore introduce an explicit *epistemic load* into the collective dynamics using relational reasoning tasks with distributed evidence. The information required to solve each task is spread across the population so that the collective contains a complete proof while individual agents initially do not. Each agent consequently carries two coupled states: a social state, represented by its current answer or vote $X_i(t)$, and an epistemic state, represented by the set of supporting facts $K_i(t)$ available to it. Local language interactions can change either or both. An agent may adopt another agent’s answer through social influence without acquiring the evidence that supports it, while an explicitly communicated fact may expand its knowledge and subsequently alter its reasoning. This distinction allows us to study how *social influence, information propagation, and reasoning jointly determine controllability*.

This separation is particularly important for feedback control. In our main setting, the controller communicates only a recommendation and does not provide task evidence. It can therefore directly perturb the social dynamics but cannot directly increase the epistemic content of the population. We can consequently distinguish whether an intervention changes what agents *believe* from whether it changes what they *know*. Moreover, by allowing the controller target to disagree with the objectively correct answer, controllability and task performance become distinct quantities: a controller can successfully steer the population while simultaneously moving it away from truth. This provides a controlled setting for asking when accumulated evidence makes a population resistant to external influence, when social pressure overwhelms reasoning, and whether the epistemic state of a collective changes the amount of control required to redirect it.

To quantify these processes, we treat sensing and actuation as two distinct information channels. Writing n_k for the current number of agents that support the controller target, Y_k for the controller’s noisy observation of that count, U_k for the action selected from that observation, and n_{k+1} for the target-support count after actuation, the two quantities are

$$I(n_k; Y_k) \quad (\text{sensing information}), \quad (2)$$

$$T_\pi(n) \equiv I(U_k; n_{k+1} \mid n_k = n) \quad (\text{action-to-population information}). \quad (3)$$

The first measures information acquired by the sensor about the current population state. The second measures, at a fixed current state, how much the controller action is expressed in the population state after the actuation step; the subscript π indicates that this channel is induced by the chosen feedback policy. We define π and the complete control loop below. These quantities are complemented by signed response statistics, population currents, and their fluctuations. We further introduce a finite-state feedback reference with one measurable directional affinity h and a kinetic compliance γ . The affinity determines the preferred direction of a controlled revision, whereas γ determines how readily a controlled opportunity produces any revision. This separation is directly calibratable from microscopic transitions and avoids introducing distinct “controller” and “epistemic-load” affinities that cannot be independently identified from the controlled channel.

This perspective connects naturally to feedback thermodynamics, where measurement and control are treated as coupled information-processing operations (Horowitz & Sandberg, 2014), and to recent stochastic-thermodynamic formulations of social imitation, where probability currents, dynamical activity, entropy production, and uncertainty relations characterize collective dynamics without requiring a literal energetic interpretation (Irisarri et al., 2026). In our finite-time feedback layer, the same single affinity leads to an exact path-space identity, $\Delta S_{\text{sys}} + hJ_c + I(n_k; Y_k) = \Sigma$, with Σ the KL irreversibility between forward and reverse feedback paths. Importantly, thermodynamics is not the motivation for introducing the controller, and Σ is not interpreted as literal energetic

dissipation of the language models. Rather, it provides a principled language for characterizing the *resource–response trade-offs* generated by the bounded feedback problem: how much information is acquired, how much directed current is produced, how irreversible the resulting coarse-grained feedback cycle is, and where additional control resources cease to be useful.

Finally, we introduce a matched finite-population q -voter process as a classical reference model. The reference system is subjected to the same population size, local interaction structure, finite sensing process, stochastic feedback policy, and bounded actuation schedule as the LLM population. It therefore captures the response expected from a system whose collective dynamics arise from stochastic local imitation alone. This comparison is deliberately not intended to claim that LLM agents are q -voters. Instead, the q -voter provides a controlled null model in which semantic communication, distributed evidence, and language-based reasoning are absent. Agreement with this reference identifies behavior that can already be explained by ordinary stochastic social influence; systematic departures reveal where the epistemic and reasoning capabilities of language-model agents modify the control dynamics.

Taken together, our goal is to develop a quantitative account of *control efficiency in reasoning multi-agent systems*: how limited sensing and limited actuation are converted into directed changes of a collective state, and how this conversion is altered when the interacting agents communicate meaningful information and reason with it. Our main contributions are:

- We formulate control of an LLM collective as a *resource-constrained feedback problem*, explicitly separating finite sensing capacity from finite actuation capacity and treating imperfect state information as a consequence of bounded observation.
- We introduce a relational multi-agent environment with distributed evidence that exposes both *social* and *epistemic* population dynamics, allowing us to distinguish changes in expressed belief from changes in the information available for reasoning.
- We develop information-theoretic and stochastic-dynamical observables that quantify sensing, transmission of controller actions, directional response, currents, fluctuations, and finite-time path irreversibility. From the same accounting we derive a bounded single-affinity thermodynamic efficiency that measures how much of the finite-time sensing–actuation expenditure is converted into directed current.
- We construct a matched finite- N q -voter reference under the same feedback protocol, providing a classical imitation baseline against which the specifically language- and reasoning-dependent behavior of the LLM population can be identified.

2 RELATED WORK

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3 BACKGROUND

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4 EXPERIMENTAL SYSTEM: DISTRIBUTED RELATIONAL REASONING

We study feedback control in a multi-agent language-model population using a synthetic relational-reasoning environment with exact ground truth and fully auditable information flow. The environment is designed to separate three ingredients that are otherwise difficult to disentangle in natural-language multi-agent tasks: (i) the underlying reasoning problem, (ii) the distribution of task-relevant information across agents, and (iii) the social dynamics through which agents exchange information and revise their decisions. Each task is generated procedurally without language-model calls, and every fact held by every agent is known exactly throughout an episode.

The central state variables for agent i at population round k are

$$X_i(k) \in \mathcal{A}, \quad K_i(k) \subseteq \mathcal{F}, \quad (4)$$

where $X_i(k)$ is the agent’s current voted relation, $\mathcal{A}$ is the finite answer set, $K_i(k)$ is the set of fact identifiers explicitly known by the agent, and $\mathcal{F}$ is the complete fact set of the task. Keeping the decision state X_i and epistemic state K_i separate is essential: two populations can have the same vote distribution while possessing very different amounts of task-relevant evidence.

4.1 SYNTHETIC RELATIONAL-REASONING TASKS

Symbolic world and ground truth. Each task is built from a symbolic two-dimensional world. Entities occupy integer coordinates, and relations are drawn from the compass-relation alphabet

$$\mathcal{R} = \{N, NE, E, SE, S, SW, W, NW\}. \quad (5)$$

A supporting chain of length L is generated as a self-avoiding sequence of exact pairwise spatial relations. If the chain contains entities $v_0, \dots, v_L$ and displacement vectors $d_1, \dots, d_L$, then the ground-truth displacement between the queried endpoints is

$$D = \sum_{\ell=1}^L d_\ell. \quad (6)$$

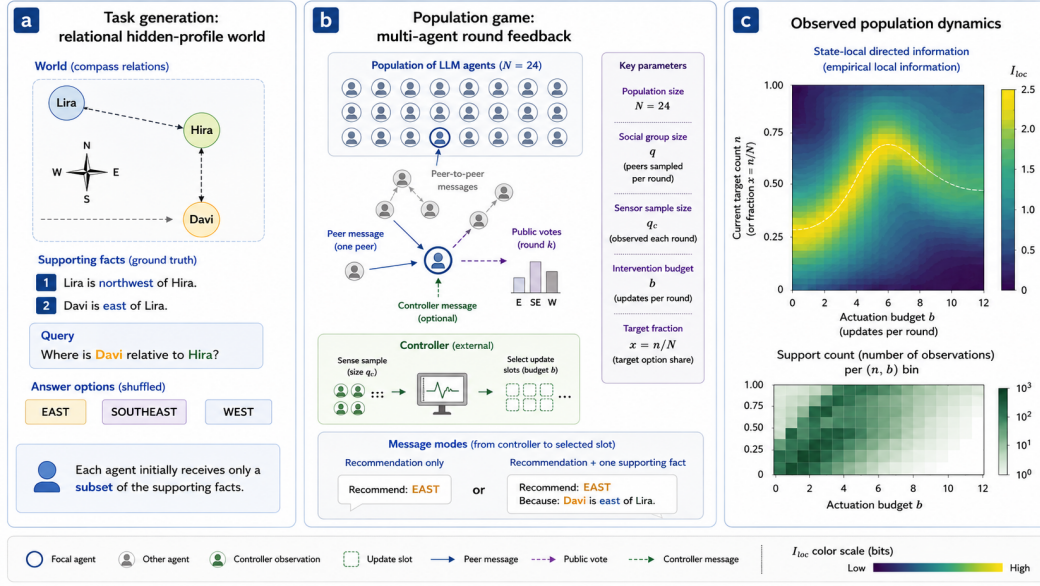


Figure 1: Overview of the multi-agent language-model control framework.

The correct answer is the compass relation induced by D . Because the symbolic coordinates are generated first, the answer is determined exactly before any natural-language rendering occurs.

Supporting facts and distractors. The L chain relations form the supporting fact set

$$\mathcal{S} = \{f_1, \dots, f_L\} \subset \mathcal{F}. \quad (7)$$

To prevent the task from reducing to trivial lexical matching, we additionally generate distractor relations in a disconnected component of the symbolic world. These distractors are internally consistent but cannot alter the query answer or shorten the required proof. The full fact set is therefore

$$\mathcal{F} = \mathcal{S} \cup \mathcal{D}, \quad (8)$$

with $\mathcal{D}$ the distractor set.

Answer options. Each task contains K distinct candidate relations, exactly one of which is correct. The persistent state of the experiment stores semantic relations rather than display labels. For every LLM call, the candidate relations are independently mapped to temporary display labels (e.g., A/B/C) using a deterministic per-call shuffle. The returned display label is immediately mapped back to its semantic relation before the state is updated. Peer votes are likewise re-rendered in the receiving agent’s current label mapping. This removes a globally stable answer-position or letter convention as a possible social attractor.

Distributed information. Each supporting fact is assigned initially to exactly r distinct agents, where r is the support redundancy. Thus, r controls how widely each proof-relevant fact is replicated across the population; it does not denote the number of agents with the complete proof. Distractors are assigned independently according to their configured redundancy.

For agent i , let $K_i(0)$ denote the initial fact set. We impose the distributed-information constraint

$$\mathcal{S} \not\subseteq K_i(0) \quad \text{for every agent } i, \quad (9)$$

while requiring

$$\mathcal{S} \subseteq \bigcup_{i=1}^N K_i(0). \quad (10)$$

Hence, no individual agent initially possesses the complete proof, whereas the population collectively does. This construction makes collective information transmission necessary while retaining exact knowledge of which evidence is available to whom.

Natural-language rendering. After the symbolic task is fixed, each relation is rendered into a canonical natural-language sentence by a deterministic template, without an LLM. The generator therefore introduces no model-dependent semantic variation into the ground truth. Each task stores the symbolic world, the natural-language facts, the answer options, the correct relation, and the exact agent–fact allocation.

Validation and reproducibility. Every generated task is validated before use. Validation checks world consistency, the required reasoning depth, uniqueness of the correct option, consistency of the rendered relations, distractor disconnection, population coverage of the proof, the no-single-agent-solution constraint, and all fact references. Task seeds are deterministic descendants of a dataset-level seed, while cryptographic fingerprints are stored for both individual tasks and the complete dataset. The dataset used in the main experiments is therefore frozen before any LLM interaction occurs.

For the principal experiments in this work we use

$$N = 24, \quad L = 2, \quad r = 6, \quad K = 3, \quad (11)$$

with four distractor facts of redundancy one. These values define the task distribution but not the controller; sensing and actuation parameters are introduced separately below.

4.2 MULTI-AGENT RELATIONAL IMITATION GAME

An episode consists of an initialization stage followed by discrete population rounds. Agents have persistent identities, votes, and explicit knowledge sets. Within a round, agents update asynchronously: after each microscopic update, the focal agent’s new vote and any newly acquired fact become immediately available to subsequent updates.

Initialization. At the beginning of an episode, each agent receives only the natural-language facts in its assigned set $K_i(0)$. Under the default *local-vote* initialization, every agent makes one independent LLM decision using only this private evidence. This produces the initial vote vector

$$\mathbf{X}(0) = (X_1(0), \dots, X_N(0)). \quad (12)$$

Because agents do not initially see the population’s other votes or private facts, this stage measures the decisions induced by the distributed evidence before social interaction begins.

Population state. Let

$$N_{j,k} = \sum_{i=1}^N \mathbb{I}[X_i(k) = a_j], \quad a_j \in \mathcal{A}, \quad (13)$$

be the number of agents voting for option a_j before round k . The complete vote-count state is

$$\mathbf{N}_k = (N_{1,k}, \dots, N_{K,k}), \quad \sum_{j=1}^K N_{j,k} = N. \quad (14)$$

When analyzing a designated controller target $Z \in \mathcal{A}$, we additionally write

$$n_{Z,k} = \sum_i \mathbb{I}[X_i(k) = Z], \quad n_k \equiv n_{Z,k}, \quad x_k = \frac{n_k}{N}. \quad (15)$$

Thus n_k is the target-support count used in the controller analysis. The scalar x_k is a useful coarse-grained population coordinate, but it is not assumed to contain the complete state of the LLM population.

One population round. At the beginning of round k , the population is described by the agents’ current votes and knowledge states,

$$\{X_i(k), K_i(k)\}_{i=1}^N.$$

The population then performs N sequential microscopic updates. At each update, one focal agent i and q peers are sampled. Importantly, only the *focal agent* generates a new LLM response; peers are

not queried again. Instead, the focal agent sees the peers’ already stored public ballots, consisting of their current vote and at most one fact that they previously chose to share.

For a focal agent i interacting with peer j , the information shown to the LLM is therefore schematically

$$(K_i(k), X_i(k), X_j(k), f_j(k)),$$

where $f_j(k)$ is either one fact already known by peer j or no shared fact. The focal agent then produces, in a single LLM call, a new ballot containing

$$(X'_i, r'_i, f'_i),$$

where X'_i is its updated vote, r'_i is a private reason, and f'_i is at most one fact from its own current knowledge that it chooses to make public.

If the peer exposes a fact f_j , that fact is permanently added to the focal agent’s knowledge,

$$K'_i = K_i \cup \{f_j\}.$$

The updated vote and public fact become the focal agent’s stored ballot and may be observed by other agents in later interactions.

External feedback controller. The game includes an external round-level controller that acts through the same social interface rather than directly overwriting an agent’s decision. Before round k , the controller samples q_c distinct agents without replacement and observes their votes only. It does not observe their knowledge sets. Based on this sensor observation it chooses an action

$$U_k \in \{\text{NO_OP}, \text{ADVOCATE}_Z\}. \quad (16)$$

If $U_k = \text{NO_OP}$, the round proceeds using ordinary peer interactions. If $U_k = \text{ADVOCATE}_Z$, exactly b of the N microscopic update positions are selected without replacement, and at each selected position one ordinary social slot is replaced by a persistent controller participant recommending target Z . Thus q_c controls sensing capacity, whereas b controls actuation capacity.

The controller is rendered as another participant rather than as an authority. In the main experiments it uses *recommendation-only* messages, meaning that it communicates a target vote but no supporting task fact. It can therefore influence decisions while leaving the epistemic state unchanged directly. This separation is important for distinguishing social steering from information transmission.

Let Y_k be the number of target supporters in the controller’s sample. The controller loop is

$$n_k \longrightarrow Y_k \xrightarrow{\pi} U_k \longrightarrow n_{k+1}. \quad (17)$$

Here Y_k is the controller’s noisy observation of the current population state, U_k is the action drawn from the feedback policy π , and n_{k+1} is the target-support count after the actuation step and the round’s asynchronous population updates. The policy is defined in Section 5.

Epistemic observables. Because every explicit fact transmission is recorded, we can measure the epistemic state of the population directly. Let $\mathcal{S}$ be the supporting fact set. We define the mean supporting-fact coverage

$$\kappa_k = \frac{1}{N} \sum_{i=1}^N \frac{|K_i(k) \cap \mathcal{S}|}{|\mathcal{S}|}, \quad (18)$$

and the fraction of agents holding the complete proof

$$\phi_k = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\mathcal{S} \subseteq K_i(k)]. \quad (19)$$

For $L = 2$, we additionally track the number of agents possessing zero, one, or two supporting facts. These variables allow us to distinguish changes in collective belief from changes in collective knowledge.

Why this environment is useful for control. The construction gives us three simultaneously observable layers: the exact symbolic task, the explicit epistemic state $\{K_i(k)\}_{i=1}^N$, and the public decision state $\{X_i(k)\}_{i=1}^N$. The controller acts only through bounded social interventions, while the population continues to exchange evidence through ordinary peer interaction. As a result, changes in population behavior can be analyzed together with the information available to agents, rather than treating the LLM population as an opaque sequence of votes.

5 EFFICIENCY OF FEEDBACK CONTROL

We study one finite feedback cycle of the coarse population coordinate rather than assigning an energetic thermodynamics to the complete language-model system. Let $n_k \in \{0, \dots, N\}$ be the number of agents supporting the controller target Z before cycle k , and let $x_k = n_k/N$. The causal order is

$$n_k \longrightarrow Y_k \longrightarrow U_k \longrightarrow n_{k+1},$$

where Y_k is a finite sensor sample and $U_k \in \{0, 1\}$ denotes NoOp or advocacy. The theory separates three questions: how strongly advocacy moves the population, how efficiently action-dependent information is converted into coherent mean motion, and how efficiently the finite-time thermodynamic budget is converted into directed current.

5.1 FINITE-INFORMATION FEEDBACK CONTROLLER

The controller observes q_c agents without replacement. If Y_k is the number of sampled agents supporting Z , then

$$S(y | n) = P(Y_k = y | n_k = n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}. \quad (20)$$

The feedback policy is

$$\pi(1 | y) = \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right], \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (21)$$

so the advocacy probability conditional on the true population state is

$$a_n \equiv P(U_k = 1 | n_k = n) = \sum_y S(y | n) \pi(1 | y). \quad (22)$$

Thus q_c , θ , and β determine when the controller acts.

5.2 SINGLE-AFFINITY FINITE-COMPLIANCE ACTUATION

A controlled microscopic opportunity is described by one directional affinity h and one kinetic compliance $\gamma \in [0, 1]$. Writing $p_h = \sigma(h)$, the count kernel is

$$K_{h,\gamma}(n+1 | n) = \gamma \frac{N-n}{N} p_h, \quad (23)$$

$$K_{h,\gamma}(n-1 | n) = \gamma \frac{n}{N} (1 - p_h), \quad (24)$$

$$K_{h,\gamma}(n | n) = 1 - K_{h,\gamma}(n+1 | n) - K_{h,\gamma}(n-1 | n). \quad (25)$$

The affinity h determines the preferred direction conditional on revision; γ determines how readily a controlled opportunity produces any revision. No second opposing affinity is introduced.

5.3 CONTROL SUSCEPTIBILITY AND EXACT RESPONSE

Conditional on the same initial state x , define the controller-induced susceptibility

$$\chi(x) \equiv \mathbb{E}[x' | U = 1, x] - \mathbb{E}[x' | U = 0, x]. \quad (26)$$

Table 1: Roles of the feedback-control quantities in the single-affinity theory.

Symbol	Role
q_c	number of agents sensed before the round
θ	feedback threshold
β	feedback-policy gain
b	controlled microscopic opportunities when active
h	directional affinity of the controlled revision channel
γ	kinetic compliance of a controlled opportunity
a_n	probability of advocacy at population state n

For the finite-compliance $q = 1$ controlled layer, b controlled opportunities give the exact response

$$\chi_{h,\gamma}(x) = [\sigma(h) - x]\Lambda_{b,\gamma}, \quad \Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b. \quad (27)$$

Hence the zero-response set point is $x^* = \sigma(h)$, while γ only rescales the response amplitude. Ordinary $q = 1$ voter updates are conditionally mean-neutral and therefore do not alter this advocacy-versus-NoOp mean response.

5.4 INFORMATION-RESPONSE EFFICIENCY

For the isolated controlled layer define

$$Q_0(m | n) = \delta_{mn}, \quad Q_1(m | n) = K_{h,\gamma}^b(m | n), \quad (28)$$

and

$$Q_\pi(m | n) = (1 - a_n)Q_0(m | n) + a_nQ_1(m | n). \quad (29)$$

The exact state-local action-to-population information is

$$T_\pi(n) \equiv I(U_k; n_{k+1} | n_k = n) \quad (30)$$

$$= \sum_{u \in \{0,1\}} P(u | n) \sum_m Q_u(m | n) \log_2 \frac{Q_u(m | n)}{Q_\pi(m | n)}. \quad (31)$$

Pinsker's inequality yields

$$T_\pi(n) \geq \frac{2a_n(1 - a_n)}{\ln 2} \chi_{h,\gamma}(n/N)^2, \quad (32)$$

which motivates the bounded information-response efficiency

$$\eta_{\text{IR}}(n) = \frac{2a_n(1 - a_n)\chi_{h,\gamma}(n/N)^2}{(\ln 2)T_\pi(n)} \leq 1. \quad (33)$$

It measures how much of the action-dependent change in the full next-state distribution is expressed as coherent mean target motion. The derivation is in Appendix A.1.

5.5 FINITE-TIME THERMODYNAMIC EFFICIENCY

The single affinity also gives a closed finite-time thermodynamic accounting. Let $p_k(n) = P(n_k = n)$ and define the coarse system entropy

$$S_{\text{sys}}[p] = - \sum_n p(n) \ln p(n) + \sum_n p(n) \ln \binom{N}{n}. \quad (34)$$

The mean target-directed controlled current is

$$J_{c,k} = N\Lambda_{b,\gamma} \sum_n p_k(n) a_n \left[\sigma(h) - \frac{n}{N} \right]. \quad (35)$$

Table 2: The two bounded efficiencies used in the feedback-control theory.

Quantity	Ratio	Interpretation
η_{IR}	coherent mean response / action information	how efficiently action-dependent distributional change becomes mean motion
η_{th}	$hJ_c / (hJ_c + I_{\text{sens}})$	how efficiently the finite-time non-storage control expenditure becomes directed current

Let $I_{\text{sens},k} \equiv I(n_k; Y_k)$ denote the information acquired by the sensor. Comparing the forward feedback path measure with its finite-time reverse reference gives the exact identity

$$\Delta S_{\text{sys},k} + hJ_{c,k} + I_{\text{sens},k} = \Sigma_k = D_{\text{KL}}\left(P_F^{(k)} \| P_R^{(k)}\right) \geq 0. \quad (36)$$

All terms in Eq. (36) use natural logarithms and are therefore measured in nats. The transient boundary term $\Delta S_{\text{sys},k}$ is retained explicitly; no stationarity assumption is required.

For target-directed operation, $hJ_{c,k} \geq 0$. Removing the transient storage term from the path irreversibility leaves the generalized finite-time control expenditure

$$C_{\text{th},k} \equiv \Sigma_k - \Delta S_{\text{sys},k} = hJ_{c,k} + I_{\text{sens},k}. \quad (37)$$

We therefore define the thermodynamic control efficiency as the fraction of this non-storage expenditure expressed as affinity-weighted directed current:

$$\eta_{\text{th},k} \equiv \frac{hJ_{c,k}}{\Sigma_k - \Delta S_{\text{sys},k}} = \frac{hJ_{c,k}}{hJ_{c,k} + I_{\text{sens},k}} \leq 1. \quad (38)$$

For $hJ_{c,k} \geq 0$ and $C_{\text{th},k} > 0$, $0 \leq \eta_{\text{th},k} \leq 1$. At stationarity, $\Delta S_{\text{sys},k} = 0$ and the same result reduces to $\eta_{\text{th}} = hJ_c / \Sigma$. Thus additional sensing lowers thermodynamic efficiency whenever it increases acquired information without a commensurate increase in directed current, whereas stronger effective actuation raises the efficiency when it increases hJ_c faster than the sensing term. The transition-ratio and path-measure calculation leading to Eqs. (36)–(38) is given in Appendix A.2.

6 RESULTS

We organize the results around three complementary questions. First, how much can the controller move the population? Second, how efficiently is the controller-dependent information expressed as coherent population motion? Third, how efficiently is the finite-time thermodynamic control budget converted into directed current? These questions correspond respectively to the susceptibility χ , the information–response efficiency η_{IR} , and the thermodynamic efficiency η_{th} .

6.1 CONTROL SUSCEPTIBILITY IS CAUSAL, FINITE, AND STATE DEPENDENT

We first measure the response to advocacy under matched initial conditions. For a fixed initial target fraction $x_0 = n_Z / N$, the susceptibility is

$$\chi(x_0) = \mathbb{E}[\Delta x \mid \text{ADV}, x_0] - \mathbb{E}[\Delta x \mid \text{NOOP}, x_0].$$

Holding the epistemic composition fixed, advocacy produces a positive shift toward the controller target at every tested state:

$$\chi(0.25) = 0.163, \quad \chi(0.375) = 0.100, \quad \chi(0.50) = 0.104.$$

The corresponding 95% matched-pair bootstrap intervals are $[0.119, 0.202]$, $[0.060, 0.144]$, and $[0.060, 0.148]$ (Fig. 2a). Thus the controller has a measurable causal influence even when the initial knowledge distribution is held fixed. The response is largest at the lowest tested target fraction and does not increase with additional pre-existing support over the tested range.

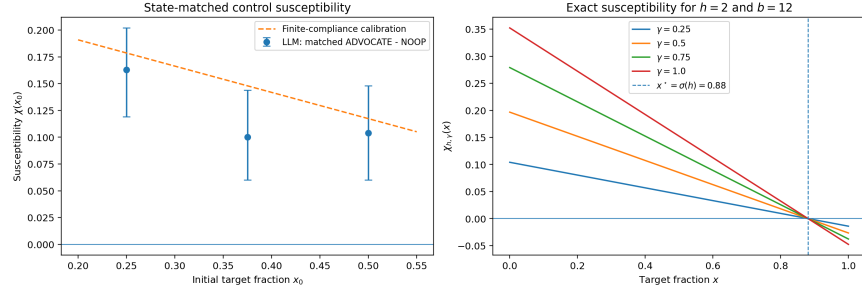


Figure 2: **Empirical and theoretical susceptibility.** (a) State-matched ADVOCATE-minus-NoOp response with 95% bootstrap intervals and the finite-compliance calibration from microscopic transition rates. (b) Exact theory showing how kinetic compliance γ changes the response amplitude without moving the intrinsic set point.

Microscopic controlled transitions provide a direct calibration of the finite-compliance description. Pooling the state-matched controlled updates gives an effective directional affinity $h_{\text{eff}} = 3.83$ and kinetic compliance $\gamma_{\text{eff}} = 0.372$. Substituting these into

$$\chi_{h,\gamma}(x) = [\sigma(h) - x] \left[1 - \left(1 - \frac{\gamma}{N} \right)^b \right]$$

gives the dashed finite-compliance reference in Fig. 2a. Because h_{eff} and γ_{eff} are estimated from the same microscopic trajectories, this is a calibrated reference rather than a parameter-free prediction. It nevertheless shows that separating directional bias from kinetic compliance removes much of the discrepancy with the idealized copying limit.

The exact theory also clarifies the role of compliance itself (Fig. 2b). Changing γ rescales the magnitude of the response while leaving the zero-response set point $x^* = \sigma(h)$ unchanged. This separation is useful empirically: h controls which direction is favored, whereas γ controls how often a controlled opportunity actually produces a revision.

6.2 STATE-LOCAL DYNAMICS REVEAL WHERE CONTROL IS EXPRESSED

The longer stochastic-feedback experiment gives access to the trajectory-level state dependence of control. Figure 3a shows the mean signed round-to-round drift $\mathbb{E}[\Delta x \mid x, b]$ for $q_c = 12$. Because the controller advocates an incorrect target, the uncontrolled epistemic dynamics usually drive x downward. Increasing b makes this correction progressively weaker, particularly at low and intermediate target support. This quantity is descriptive rather than causal: the matched experiment above remains the clean measurement of χ .

The controller action also leaves a strongly state-dependent information signature. We measure the reduced-state conditional mutual information

$$T_\pi(n) \equiv I(U_k; n_{k+1} \mid n_k = n),$$

the action-to-population information induced by the chosen feedback policy π . Here U_k is the policy-selected controller action and n_{k+1} is the population state after actuation. The empirical excess above a within-state permutation null is shown in Fig. 3b. At small b the signal is weak, whereas at $b = 18$ a clear interior band appears. The exact finite-state theory produces the same basic geometric mechanism (Fig. 3c): local information is high only where the action is not already deterministic and where the advocacy and NoOp transition kernels are sufficiently separated. The support map in Fig. 3d is included because the higher state bins are visited less often and should not be interpreted with the same confidence as densely sampled cells.

The robust conclusion is that control information is not uniform over population state space and stronger actuation creates an informative interior region. We do not interpret the precise ridge location as a phase boundary because n_Z is only a coarse-grained state.

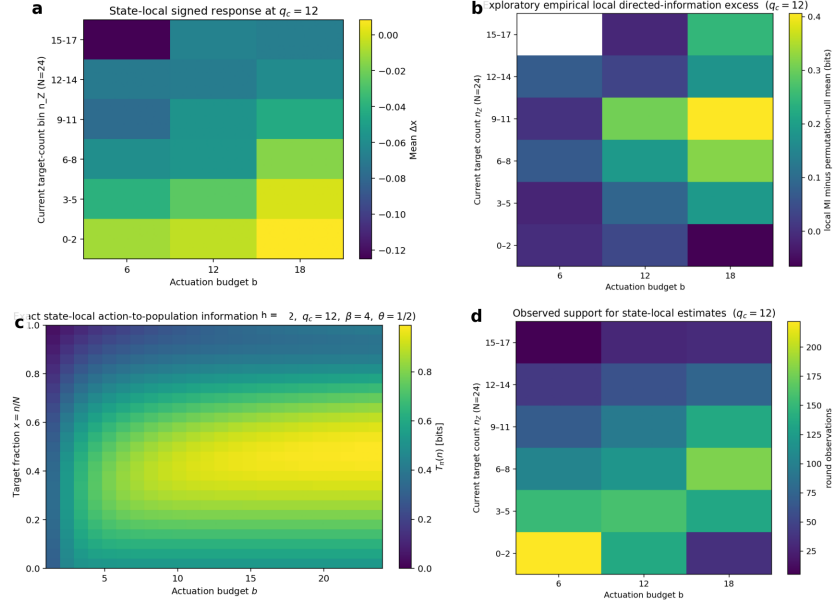


Figure 3: **State-local response and information.** (a) Empirical signed population drift as a function of target support and actuation budget at $q_c = 12$. (b) Empirical state-local action-to-population information above a within-state permutation null. (c) Exact theoretical state-local information landscape for the baseline single-affinity finite-compliance feedback controller. (d) Number of observed rounds supporting each empirical state bin.

6.3 FEEDBACK GAIN CONCENTRATES RATHER THAN SIMPLY INCREASES INFORMATION

The theory makes a more specific prediction about the feedback policy. The gain β controls how sharply the controller switches between advocacy and NoOp after sensing. At small β , the action remains uncertain over a wide range of states; at large β , the policy becomes nearly deterministic except close to its switching threshold. Consequently increasing β does not monotonically increase $T_\pi(n)$. Instead, the informative band narrows and moves toward the decision boundary (Fig. 4a).

The state-averaged theory curves make the same point in one dimension (Fig. 4b): for fixed actuation budget, larger β can reduce the average action-to-population information even though the controller is more decisive. This separates controller *decisiveness* from controller *information transmission* and suggests a focused validation rather than a broad search: $\beta \in \{1, 4, 8\}$ is sufficient to test whether the empirical information band sharpens as predicted.

6.4 INFORMATION-RESPONSE EFFICIENCY INCREASES WITH ACTUATION

The information-response efficiency asks how much of the controller-dependent change in the full next-state distribution is expressed as a coherent shift of the mean target fraction. Using the bounded quantity defined in Section 5, the occupancy-averaged empirical values are small in absolute terms but systematic across the resource grid (Fig. 5a). They range from approximately 0.4% to 2.5% of the rigorous Pinsker bound and increase with b for every sensing budget. At $q_c = 12$, for example,

$$\eta_{\text{IR}} = 0.0043, 0.0135, 0.0228 \quad \text{for } b = 6, 12, 18.$$

Thus stronger actuation does more than increase the size of the population response: within this operating regime it also makes a larger fraction of the action-dependent distributional change visible as coherent mean motion. The small absolute values should not be interpreted as energetic losses; the Pinsker inequality supplies a conservative information-theoretic ceiling and need not be tight for these transition distributions.

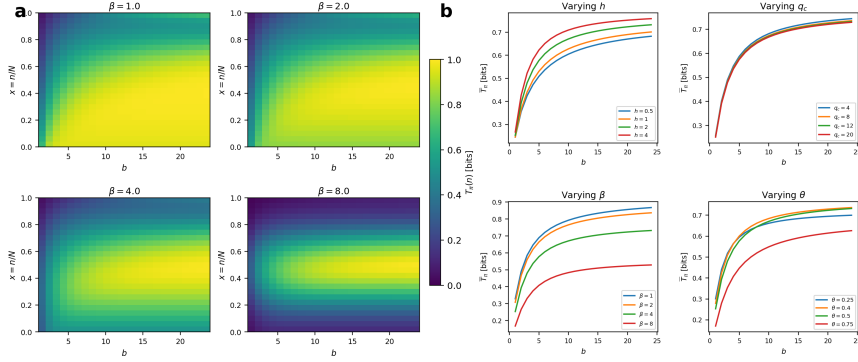


Figure 4: **Exact theory for feedback gain.** (a) State-local action-to-population information for increasing policy gain β . A sharper policy concentrates information around the switching region and suppresses it where the action saturates. (b) State-averaged local information versus actuation budget for the theory parameter sweeps; the β family shows that a more deterministic policy need not transmit more action information overall.

6.5 THERMODYNAMIC EFFICIENCY EXPOSES THE SENSING–ACTUATION TRADE-OFF

The single-affinity path identity leads to a bounded thermodynamic control efficiency without introducing an opposing load. Removing the transient system entropy term from the finite-time path irreversibility gives

$$\Sigma - \Delta S_{\text{sys}} = hJ_c + I(n_k; Y_k),$$

and therefore

$$\eta_{\text{th}} = \frac{hJ_c}{\Sigma - \Delta S_{\text{sys}}} = \frac{hJ_c}{hJ_c + I(n_k; Y_k)} \leq 1.$$

The numerator is the affinity-weighted target-directed current; the denominator is the non-storage part of the finite-time thermodynamic accounting. Thus η_{th} asks what fraction of the sensing-plus-actuation expenditure is expressed as directed population motion. Appendix A.2 gives the complete path-measure derivation.

Figure 5b shows the exact single-affinity prediction for the same sensing budgets used experimentally, with $h = 2$, $\gamma = 0.75$, $\beta = 4$, $\theta = 1/2$, and $p_0(n) = \text{Binomial}(24, 0.25)$. At fixed q_c , the efficiency rises with actuation because the controlled current grows while the sensing information is fixed by the sensor channel. At fixed b , larger sensing budgets lower the efficiency unless the additional information produces a commensurate increase in current. For $b = 18$, for example,

$$\eta_{\text{th}} = 0.985, 0.964, 0.930 \quad \text{for } q_c = 6, 12, 18.$$

The result is therefore not that sensing is intrinsically undesirable, but that information that does not change the action-relevant population response is a thermodynamic overhead in this control objective.

The empirical sensing measurements support precisely this mechanism (Fig. 5c). Increasing q_c substantially raises $I(n_k; Y_k)$, whereas the calibrated controlled current changes much less across the same resource grid. Combined with the close agreement between the measured current and the finite-compliance current reported above, this yields a direct design prediction: in the present regime, adding actuation improves both η_{IR} and η_{th} , whereas additional sensing is efficient only when the policy can convert it into additional directed current.

Taken together, the results support a single resource-conversion picture of feedback control. Susceptibility identifies where the population can be moved; state-local information identifies where the action is dynamically visible; η_{IR} quantifies how coherently that information appears as mean motion; and η_{th} quantifies how efficiently the finite-time sensing–actuation budget is converted into affinity-weighted directed current. In the present regime, actuation is the dominant useful resource, while sensing displays clear diminishing returns. The theory additionally predicts that feedback gain β should reshape, rather than simply amplify, the information channel, providing a focused next validation of the framework.

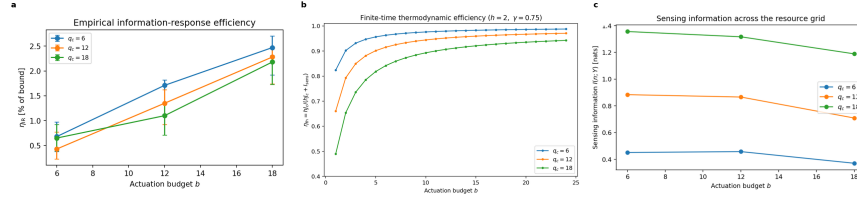


Figure 5: **The two control efficiencies and their resource dependence.** (a) Empirical information-response efficiency across the sensing-actuation grid. (b) Exact single-affinity thermodynamic efficiency $\eta_{\text{th}} = hJ_c / (hJ_c + I_{\text{sens}})$ as the actuation budget is varied for the three sensing budgets used experimentally. (c) Empirical sensing information, which rises strongly with q_c without a proportional increase in controlled current.

7 CONCLUSION

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A THEORY AND DERIVATIONS

A.1 DERIVATION OF THE INFORMATION-RESPONSE BOUND

This appendix derives the bound

$$T_{\pi}(n) \geq \frac{2a_n(1-a_n)}{\ln 2} \chi_{h,\gamma}(n/N)^2 \quad (39)$$

used to define the information-response efficiency. The derivation proceeds in three steps: (i) expressing the conditional mutual information as a weighted Jensen-Shannon divergence, (ii) applying Pinsker’s inequality, and (iii) bounding the change in the mean target fraction by total variation.

Throughout the derivation, the current population state n is held fixed.

A.1.1 CONDITIONAL MUTUAL INFORMATION AS A JENSEN–SHANNON DIVERGENCE

Let the binary control variable be

$$U \in \{0, 1\},$$

where $U = 0$ denotes no intervention and $U = 1$ denotes advocacy of the controller target Z . At population state n , define

$$\Pr(U = 1 \mid n) = a_n, \quad \Pr(U = 0 \mid n) = 1 - a_n. \quad (40)$$

The corresponding next-round target-count distributions are

$$Q_0(n' \mid n) = \Pr(n' \mid U = 0, n), \quad (41)$$

$$Q_1(n' \mid n) = \Pr(n' \mid U = 1, n). \quad (42)$$

For notational convenience, while n is fixed, write

$$P(n') \equiv Q_0(n' \mid n), \quad Q(n') \equiv Q_1(n' \mid n), \quad a \equiv a_n. \quad (43)$$

The conditional mutual information between the control action and the next population state is, in the round-indexed notation of the main text,

$$T_\pi(n) \equiv I(U_k; n_{k+1} \mid n_k = n).$$

Holding $n_k = n$ fixed, we use the shorthand $U \equiv U_k$ and $n' \equiv n_{k+1}$ below. Thus

$$I(U; n' \mid n) = \sum_{u, n'} \Pr(u, n' \mid n) \log_2 \frac{\Pr(u, n' \mid n)}{\Pr(u \mid n) \Pr(n' \mid n)}. \quad (44)$$

Using the factorization

$$\Pr(u, n' \mid n) = \Pr(u \mid n) \Pr(n' \mid u, n),$$

the factor $\Pr(u \mid n)$ cancels inside the logarithm:

$$\begin{aligned} I(U; n' \mid n) &= \sum_{u, n'} \Pr(u \mid n) \Pr(n' \mid u, n) \log_2 \frac{\Pr(n' \mid u, n)}{\Pr(n' \mid n)} \\ &= \sum_u \Pr(u \mid n) D_{\text{KL}}(\Pr(n' \mid u, n) \parallel \Pr(n' \mid n)). \end{aligned} \quad (45)$$

We therefore need the next-state distribution when the value of U is not specified. By the law of total probability,

$$\begin{aligned} \Pr(n' \mid n) &= \sum_{u \in \{0, 1\}} \Pr(u \mid n) \Pr(n' \mid u, n) \\ &= (1 - a)P(n') + aQ(n'). \end{aligned} \quad (46)$$

Define the mixture distribution

$$M(n') \equiv (1 - a)P(n') + aQ(n'). \quad (47)$$

Substituting the two possible actions into Eq. equation 45 gives

$$I(U; n' \mid n) = (1 - a)D_{\text{KL}}(P \parallel M) + aD_{\text{KL}}(Q \parallel M). \quad (48)$$

The weighted Jensen–Shannon divergence is defined by

$$\text{JS}_a(P, Q) \equiv (1 - a)D_{\text{KL}}(P \parallel M) + aD_{\text{KL}}(Q \parallel M), \quad M = (1 - a)P + aQ. \quad (49)$$

Hence, exactly,

$$I(U; n' \mid n) = \text{JS}_a(P, Q) \quad (50)$$

or, restoring the round-indexed notation,

$$T_\pi(n) \equiv I(U_k; n_{k+1} \mid n_k = n) = \text{JS}_{a_n}[Q_0(\cdot \mid n), Q_1(\cdot \mid n)]. \quad (51)$$

Up to this point, all relations are exact; no bound has yet been used.

A.1.2 PINSKER BOUND ON THE JENSEN–SHANNON DIVERGENCE

For two probability distributions A and B , define their total variation distance as

$$\text{TV}(A, B) \equiv \frac{1}{2} \sum_x |A(x) - B(x)|. \quad (52)$$

Pinsker’s inequality states that, when the Kullback–Leibler divergence is measured in bits,

$$D_{\text{KL}}(A\|B) \geq \frac{2}{\ln 2} \text{TV}(A, B)^2. \quad (53)$$

The factor $1/\ln 2$ appears because the standard Pinsker inequality is conventionally written for natural logarithms, whereas here the information is reported in bits.

Applying Eq. equation 53 separately to the two KL terms in Eq. equation 48 yields

$$T_\pi(n) \geq \frac{2}{\ln 2} [(1-a) \text{TV}(P, M)^2 + a \text{TV}(Q, M)^2]. \quad (54)$$

The two total variation distances simplify because M is a mixture of P and Q . From

$$M - P = (1-a)P + aQ - P = a(Q - P),$$

we obtain

$$\begin{aligned} \text{TV}(P, M) &= \frac{1}{2} \sum_{n'} |a(Q(n') - P(n'))| \\ &= a \text{TV}(P, Q). \end{aligned} \quad (55)$$

Similarly,

$$Q - M = Q - (1-a)P - aQ = (1-a)(Q - P),$$

and therefore

$$\text{TV}(Q, M) = (1-a) \text{TV}(P, Q). \quad (56)$$

Substituting Eqs. equation 55–equation 56 into Eq. equation 54,

$$\begin{aligned} T_\pi(n) &\geq \frac{2}{\ln 2} [(1-a)a^2 + a(1-a)^2] \text{TV}(P, Q)^2 \\ &= \frac{2}{\ln 2} a(1-a) \underbrace{[a + (1-a)]}_{=1} \text{TV}(P, Q)^2. \end{aligned} \quad (57)$$

Thus

$$T_\pi(n) \geq \frac{2a(1-a)}{\ln 2} \text{TV}(P, Q)^2. \quad (58)$$

Restoring $a = a_n$, $P = Q_0$, and $Q = Q_1$,

$$T_\pi(n) \geq \frac{2a_n(1-a_n)}{\ln 2} \text{TV}(Q_0(\cdot | n), Q_1(\cdot | n))^2. \quad (59)$$

A.1.3 FROM TOTAL VARIATION TO THE MEAN TARGET-FRACTION RESPONSE

Let $n' \in \{0, \dots, N\}$ denote the next-round number of agents supporting the controller target. Define the corresponding target fraction

$$f(n') \equiv \frac{n'}{N}. \quad (60)$$

Since $0 \leq n' \leq N$,

$$0 \leq f(n') \leq 1. \quad (61)$$

Define the intervention-induced change in the mean target fraction by

$$\begin{aligned}\chi_{h,\gamma}(n/N) &\equiv \mathbb{E}_{Q_1} \left[\frac{n'}{N} \middle| n \right] - \mathbb{E}_{Q_0} \left[\frac{n'}{N} \middle| n \right] \\ &= \mathbb{E}_Q[f] - \mathbb{E}_P[f].\end{aligned}\tag{62}$$

For completeness, we now show that any function satisfying $0 \leq f \leq 1$ obeys

$$|\mathbb{E}_Q[f] - \mathbb{E}_P[f]| \leq \text{TV}(P, Q).\tag{63}$$

Define

$$\Delta(n') \equiv Q(n') - P(n').$$

Since both P and Q are normalized,

$$\sum_{n'} \Delta(n') = 0.\tag{64}$$

Let

$$A_+ = \{n' : \Delta(n') \geq 0\}.$$

Then

$$\sum_{n' \in A_+} \Delta(n') = \text{TV}(P, Q).\tag{65}$$

Using Eq. equation 62,

$$\begin{aligned}\chi_{h,\gamma}(n/N) &= \sum_{n'} f(n') \Delta(n') \\ &= \sum_{n' \in A_+} f(n') \Delta(n') + \sum_{n' \notin A_+} f(n') \Delta(n').\end{aligned}\tag{66}$$

The second sum is non-positive because $f(n') \geq 0$ and $\Delta(n') < 0$ outside A_+ . Therefore

$$\begin{aligned}\chi_{h,\gamma}(n/N) &\leq \sum_{n' \in A_+} f(n') \Delta(n') \\ &\leq \sum_{n' \in A_+} \Delta(n') \\ &= \text{TV}(P, Q),\end{aligned}\tag{67}$$

where the second inequality uses $f(n') \leq 1$. Repeating the argument after exchanging P and Q gives the corresponding lower bound. Hence

$$|\chi_{h,\gamma}(n/N)| \leq \text{TV}(P, Q).\tag{68}$$

Squaring both sides,

$$\chi_{h,\gamma}(n/N)^2 \leq \text{TV}(P, Q)^2.\tag{69}$$

Finally, combining Eqs. equation 58 and equation 69,

$$\begin{aligned}T_\pi(n) &\geq \frac{2a_n(1-a_n)}{\ln 2} \text{TV}(Q_0(\cdot | n), Q_1(\cdot | n))^2 \\ &\geq \frac{2a_n(1-a_n)}{\ln 2} \chi_{h,\gamma}(n/N)^2.\end{aligned}\tag{70}$$

Therefore,

$$T_\pi(n) \geq \frac{2a_n(1-a_n)}{\ln 2} \chi_{h,\gamma}(n/N)^2.\tag{71}$$

For reference, the complete chain of relations is

$$\begin{aligned}
 T_\pi(n) &= I(U_k; n_{k+1} | n_k = n) \\
 &= \text{JS}_{a_n} [Q_0(\cdot | n), Q_1(\cdot | n)] \\
 &\geq \frac{2a_n(1-a_n)}{\ln 2} \text{TV}(Q_0(\cdot | n), Q_1(\cdot | n))^2 \\
 &\geq \frac{2a_n(1-a_n)}{\ln 2} \chi_{h,\gamma}(n/N)^2.
 \end{aligned} \tag{72}$$

A.2 DERIVATION OF THE THERMODYNAMIC EFFICIENCY

This appendix derives the finite-time identity and the bounded thermodynamic control efficiency stated in Section 5.5. The calculation uses only the single directional affinity h , the kinetic compliance γ , and the finite sensor-policy channel.

A.2.1 CONTROLLED TRANSITION RATIO

For one controlled microscopic opportunity,

$$K_{h,\gamma}(n+1 | n) = \gamma \frac{N-n}{N} \sigma(h), \quad K_{h,\gamma}(n | n+1) = \gamma \frac{n+1}{N} \sigma(-h).$$

Since $\sigma(h)/\sigma(-h) = e^h$,

$$\frac{K_{h,\gamma}(n+1 | n)}{K_{h,\gamma}(n | n+1)} = \frac{N-n}{n+1} e^h. \tag{73}$$

The compliance factor cancels exactly: γ changes how frequently controlled revisions occur, but not their forward/reverse directional ratio. Define the combinatorial macrostate entropy

$$S_{\text{mix}}(n) = \ln \binom{N}{n}. \tag{74}$$

Because

$$S_{\text{mix}}(n+1) - S_{\text{mix}}(n) = \ln \frac{N-n}{n+1},$$

Eq. (73) becomes

$$\ln \frac{K_{h,\gamma}(n+1 | n)}{K_{h,\gamma}(n | n+1)} = S_{\text{mix}}(n+1) - S_{\text{mix}}(n) + h. \tag{75}$$

Equivalently, the positive weight

$$w_h(n) = \binom{N}{n} e^{hn} \tag{76}$$

satisfies

$$w_h(n) K_{h,\gamma}(m | n) = w_h(m) K_{h,\gamma}(n | m) \tag{77}$$

for every supported transition. Weighted reversibility is preserved under matrix multiplication. Hence for the b -opportunity advocacy kernel $Q_1 = K_{h,\gamma}^b$,

$$\ln \frac{Q_1(m | n)}{Q_1(n | m)} = S_{\text{mix}}(m) - S_{\text{mix}}(n) + h(m-n). \tag{78}$$

For NoOp, $Q_0(m | n) = \delta_{mn}$.

A.2.2 FORWARD AND REVERSE FEEDBACK PATHS

At feedback cycle k , let $p_k(n) = P(n_k = n)$. The forward path measure on (n, y, u, m) is

$$P_F^{(k)}(n, y, u, m) = p_k(n) S(y | n) \pi(u | y) Q_u(m | n). \tag{79}$$

Its final population marginal is

$$p_{k+1}(m) = \sum_{n,y,u} P_F^{(k)}(n, y, u, m), \quad (80)$$

and the forward sensor marginal is

$$p_{Y,k}(y) = \sum_n p_k(n) S(y | n). \quad (81)$$

We use the finite-time reverse reference

$$P_R^{(k)}(m, y, u, n) = p_{k+1}(m) p_{Y,k}(y) \pi(u | y) Q_u(n | m). \quad (82)$$

The reverse reference starts from the forward final ensemble p_{k+1} and therefore does not impose stationarity. The same policy factor $\pi(u | y)$ appears in both path weights and cancels from their ratio.

Define the stochastic path irreversibility

$$\Sigma_k^* \equiv \ln \frac{P_F^{(k)}(n, y, u, m)}{P_R^{(k)}(m, y, u, n)}. \quad (83)$$

Substitution gives

$$\Sigma_k^* = \ln \frac{p_k(n)}{p_{k+1}(m)} + \ln \frac{S(y | n)}{p_{Y,k}(y)} + \ln \frac{Q_u(m | n)}{Q_u(n | m)}. \quad (84)$$

Introduce the pointwise sensing information

$$i_k(n; y) = \ln \frac{S(y | n)}{p_{Y,k}(y)}, \quad (85)$$

the stochastic system entropy

$$s_{\text{sys}}(n; p) = -\ln p(n) + S_{\text{mix}}(n), \quad (86)$$

and the controlled target-count current increment

$$j_c = u(m - n). \quad (87)$$

For $u = 0$, $m = n$ and $j_c = 0$. For $u = 1$, Eq. (78) applies. Therefore

$$\Sigma_k^* = \Delta s_{\text{sys}} + h j_c + i_k(n; y). \quad (88)$$

A.2.3 FINITE-TIME SECOND LAW AND MEAN CONTROLLED CURRENT

Averaging Eq. (88) over the forward ensemble gives

$$\Sigma_k \equiv \mathbb{E}_F[\Sigma_k^*] = D_{\text{KL}}(P_F^{(k)} \| P_R^{(k)}) \geq 0. \quad (89)$$

The three average terms are

$$\Delta S_{\text{sys},k} = S_{\text{sys}}[p_{k+1}] - S_{\text{sys}}[p_k], \quad (90)$$

$$J_{c,k} = \mathbb{E}_F[j_c], \quad (91)$$

$$I_{\text{sens},k} = I(n_k; Y_k) = \sum_{n,y} p_k(n) S(y | n) \ln \frac{S(y | n)}{p_{Y,k}(y)}, \quad (92)$$

where

$$S_{\text{sys}}[p] = -\sum_n p(n) \ln p(n) + \sum_n p(n) \ln \binom{N}{n}. \quad (93)$$

Thus

$$\Delta S_{\text{sys},k} + h J_{c,k} + I_{\text{sens},k} = \Sigma_k \geq 0. \quad (94)$$

The current can be obtained directly from the susceptibility. Conditional on $n_k = n$ and advocacy,

$$\mathbb{E}[m - n | n, U = 1] = N \chi_{h,\gamma}(n/N) = N \Lambda_{b,\gamma} \left[\sigma(h) - \frac{n}{N} \right].$$

Advocacy occurs with probability a_n , so

$$J_{c,k} = N \Lambda_{b,\gamma} \sum_n p_k(n) a_n \left[\sigma(h) - \frac{n}{N} \right]. \quad (95)$$

This is the same current entering the affinity term of Eq. (94).

A.2.4 THERMODYNAMIC CONTROL EFFICIENCY

Equation (94) contains a finite-time boundary term, $\Delta S_{\text{sys},k}$, because the population ensemble need not be stationary. To isolate the part of path irreversibility associated with the feedback operation itself, define the non-storage control expenditure

$$C_{\text{th},k} \equiv \Sigma_k - \Delta S_{\text{sys},k}. \quad (96)$$

Using Eq. (94),

$$C_{\text{th},k} = hJ_{c,k} + I_{\text{sens},k}. \quad (97)$$

The two contributions have distinct operational meanings. The first is the affinity-weighted directed population current produced by actuation; the second is the information acquired by sensing before the action is chosen.

For target-directed operation, $hJ_{c,k} \geq 0$. The fraction of the non-storage control expenditure expressed as directed current is therefore

$$\eta_{\text{th},k} \equiv \frac{hJ_{c,k}}{C_{\text{th},k}} = \frac{hJ_{c,k}}{\Sigma_k - \Delta S_{\text{sys},k}} = \frac{hJ_{c,k}}{hJ_{c,k} + I_{\text{sens},k}}. \quad (98)$$

Since $I_{\text{sens},k} \geq 0$,

$$0 \leq \eta_{\text{th},k} \leq 1 \quad (hJ_{c,k} \geq 0, C_{\text{th},k} > 0). \quad (99)$$

At stationarity the boundary term vanishes and

$$\eta_{\text{th}} = \frac{hJ_c}{\Sigma}. \quad (100)$$

The finite-time expression is therefore not obtained by silently assuming $\Delta S_{\text{sys}} = 0$; rather, it removes that transient storage term explicitly before forming the ratio.

The efficiency makes the sensing–actuation trade-off transparent. Holding the sensor channel fixed, increasing actuation can raise hJ_c and drive η_{th} toward one. Conversely, increasing q_c can increase I_{sens} without a proportional increase in J_c , thereby reducing η_{th} . Kinetic compliance enters through J_c : a controller can have a large directional affinity but low thermodynamic efficiency when agents rarely comply and the resulting current remains small relative to the sensing information acquired.

B RELATIONAL-REASONING PROMPT

The population is driven by one versioned prompt family, `relational_public_ballot` version 1. Each call is compiled into two chat messages. The system message establishes the agent’s role, privacy boundary, task, call-specific option mapping, and currently known facts. The user message supplies the current vote and visible social source, then states the response contract. The model therefore receives only the state available to the focal agent at that microscopic update; it is never shown the global vote vector, other agents’ private facts, the ground-truth answer, or the controller’s sensor sample.

The display below reproduces one complete prompt from the audit of the minimal Study 04 inspection episode ($N = 24$, $q_c = 12$, $b = 12$). It is the first focal update of population round 1 for Agent 17. Typography is annotation only: **orange** marks values instantiated anew for the call, **green** marks the focal agent’s private state, and **blue** marks the visible social input. None of these colours or the “SYSTEM MESSAGE”/“USER MESSAGE” labels are sent to the model.

Audited focal-update prompt

SYSTEM MESSAGE

You are **Agent 17**, one participant in a group reasoning problem.

Your goal is to identify the correct answer.

Participant identities persist across interactions.

Different participants know different facts about this problem. No single participant is guaranteed to know everything needed to answer, and some participants may have objectives that differ from yours. You do not know which participants, if any, have different objectives.

Make your own decision, using:

- the facts you currently know;
- your current position;
- the public positions of the participants shown below.

You see only the participants shown in this interaction, and of each one you see their vote and the fact they chose to share, if any.

Other participants will see the same of you: your vote and any fact you choose to share, and nothing else.

Your reason is your own record: it is not shown to anyone.

QUESTION

Where is Davi relative to Hira?

The available answers are:

- A) WEST
- B) EAST
- C) SOUTHWEST

Exactly one of these answers is correct. Vote by its letter.

YOUR CURRENT KNOWLEDGE

You do not currently know any facts about this problem. You will have to rely on what other participants tell you.

USER MESSAGE

YOUR CURRENT POSITION

Vote: SOUTHWEST

CURRENT SOCIAL INFORMATION

Agent 25

Vote: B

DECISION

Work out which option the facts available to you support, and vote for it.

Your reason should briefly explain your choice, for your own record.

Sharing a fact is the only way to pass information to other participants. You may share exactly one of the facts you currently know by giving its identifier in `shared_fact_id`, so that the participants who see your position can use it too. Use “none” if you prefer to share nothing.

You may share only a fact listed under YOUR CURRENT KNOWLEDGE, by its exact identifier. Do not invent facts, identifiers, or relationships that were not given to you.

Keep your reason to at most three sentences.

Return only valid JSON:

```
{
  "vote": "<A | B | C>",
  "reason": "<brief private reason>",
  "shared_fact_id": "<none>"
}
```

What changes between calls. The identity, query, available semantic answers, private facts, current vote, visible sources, and admissible fact identifiers are bound from the live game state. The semantic answers are independently permuted onto A/B/C for every call. Thus the `social vote B` above means *EAST* only in this instance; the persistent population state stores EAST, not the letter B. This prevents a stable display position from becoming an unintended convention.

Ordinary and controlled social inputs. An uncontrolled update shows the vote and, when supplied, one valid fact from an ordinary peer. In the audited example the visible source is the controller: the controller’s semantic target EAST is rendered as `Vote: B` under this call’s permutation. The prompt deliberately does not identify that source as a controller or authority. Under the `recommendation_only` mode it also supplies no fact, so the intervention can change the public vote state but cannot directly enlarge the focal agent’s knowledge set.

Initialization variant. Before social interaction, each of the 24 agents receives the same role, task, privacy, fact-sharing, and JSON-contract blocks, but no current-position or social-information block. Instead, the prompt states that no other participant has yet declared a position and instructs the agent to decide from its local facts alone. The resulting 24 ballots define the initial population state.

Response and information boundary. The accepted response contains a display-letter vote, a private reason of at most three sentences, and either one fact identifier already in the agent’s knowledge set or `none`. The display letter is immediately mapped back to its semantic relation. The reason is retained for audit but never shown to another agent. Only the semantic vote and the explicitly selected fact may enter a later prompt, which makes the public decision channel and the epistemic transmission channel separately observable.

Audit check. The inspection episode recorded 72 distinct compiled prompt instances: 24 local-initialization prompts and 48 focal-update prompts (24 in each of two population rounds). All 72 responses passed validation without a retry or provider error. The prompt definition hash was constant across the episode, whereas every instance hash was unique, confirming that one fixed template was bound to 72 call-specific views.

C CONTROLLED SUSCEPTIBILITY STUDY

C.1 STATE-MATCHED RESPONSE TO ADVOCACY

We isolate the one-round behavioral response of the population to controller advocacy while holding the epistemic population fixed. The experiment uses $N = 24$ agents, $q = 1$ social interaction, reasoning depth $L = 2$, support redundancy $r = 6$, recommendation-only control, and an incorrect controller target Z . We fix the sensing and actuation budgets to $q_c = b = 18$ and explicitly initialize three target-support levels,

$$x_0 \in \{0.25, 0.375, 0.50\}, \quad n_Z(0) \in \{6, 9, 12\}.$$

Within each task, agent identities and initial knowledge sets $K_i(0)$ are identical across all three conditions, giving $\kappa_0 = 0.25$ and $\phi_0 = 0$. Target supporters are selected deterministically and balanced across the populated knowledge strata, with nested sets $Z_6 \subset Z_9 \subset Z_{12}$; all remaining agents are initialized at the correct relation. Thus only the initial social support for Z changes across conditions.

For each x_0 , we compare two deterministic one-round interventions: **ADVOCATE**, in which the controller advocates for Z at exactly $b = 18$ controlled update positions, and **NOOP**, in which it does not intervene. The two arms use matched episode seeds. We evaluate two tasks with ten matched repetitions per task, yielding 120 one-round episodes. The sensor remains active for logging, but it does not determine the forced action.

The primary quantity is the signed intervention response

$$\chi(x_0) = \mathbb{E}[x_1 - x_0 \mid \text{ADV}, x_0] - \mathbb{E}[x_1 - x_0 \mid \text{NOOP}, x_0], \quad (101)$$

with 95% confidence intervals from 1000 bootstrap resamples of matched episode pairs. Since the action is deterministic within each arm, this experiment estimates a response contrast rather than an action-to-population mutual information.

The measured responses are $\chi(0.25) = 0.163$ (95% CI $[0.119, 0.202]$), $\chi(0.375) = 0.100$ ($[0.060, 0.144]$), and $\chi(0.50) = 0.104$ ($[0.060, 0.148]$). The controller therefore produces a positive causal shift toward its target at all three matched states. However, the largest response occurs at the lowest tested target fraction, and increasing the initial target support from $x_0 = 0.375$ to 0.50 produces no additional gain. The exact $q = 1$ reference predicts the same qualitative ordering—largest leverage at low target support—but substantially larger responses (0.401, 0.334, and 0.268, respectively). Thus, in this controlled one-round setting, LLM advocacy is effective but markedly weaker than the idealized copying reference, and there is no evidence that additional pre-existing social support amplifies the intervention response.

C.2 EMPIRICAL CALIBRATION OF DIRECTIONAL AFFINITY AND COMPLIANCE

The single-affinity theory describes a controlled interaction using a directional bias h and a kinetic compliance γ . To connect these quantities to the LLM dynamics, we use the microscopic transitions observed in the state-matched intervention experiment. We coarse-grain each focal agent’s vote into two states: supporting the controller target Z , or not supporting Z . Among controlled updates, define

$$p_+ = P(\text{non-}Z \rightarrow Z \mid \text{controlled}), \quad p_- = P(Z \rightarrow \text{non-}Z \mid \text{controlled}).$$

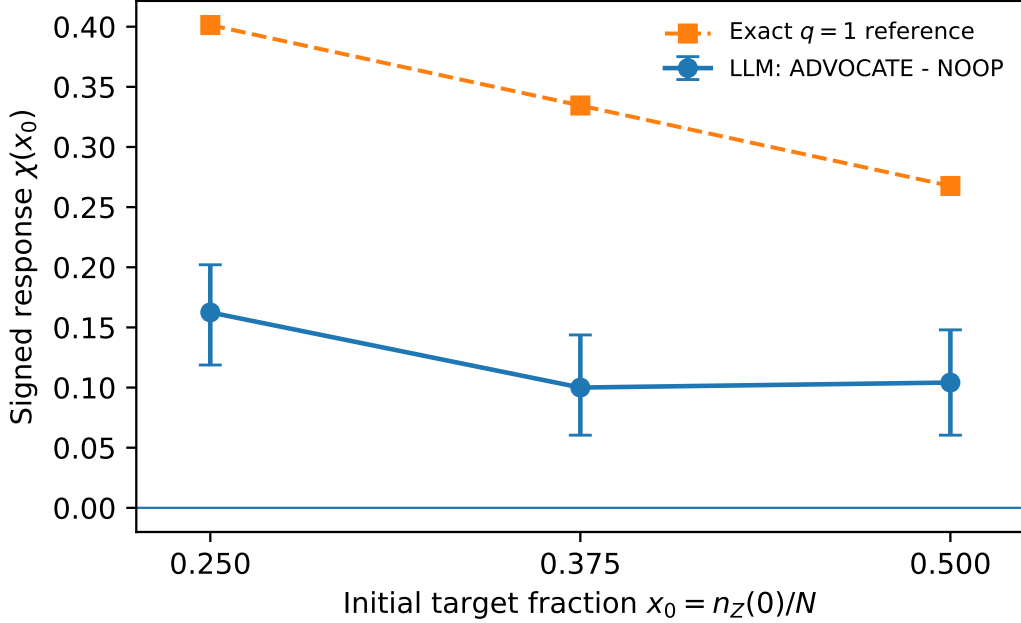


Figure 6: One-round state-matched response to advocacy. Points show the empirical ADVOCATE-minus-NOOP response with 95% matched-pair bootstrap intervals. The dashed curve is the exact $q = 1$ reference at $N = 24$ and $b = 18$, $\chi_{q=1}(x_0) = (1 - x_0)[1 - (1 - 1/N)^b]$. Advocacy produces a positive response at every imposed initial target fraction, while the response does not increase with pre-existing target support over the tested range.

Table 3: Microscopic transitions observed at controlled update positions.

Transition	Eligible updates	Observed transitions	Probability
non- $Z \rightarrow Z$	572	208	$p_+ = 0.364$
$Z \rightarrow$ non- Z	508	4	$p_- = 0.0079$

Under the finite-compliance kernel,

$$p_+ = \gamma\sigma(h), \quad p_- = \gamma\sigma(-h).$$

Therefore the two parameters are directly identifiable:

$$\gamma_{\text{eff}} = p_+ + p_-, \quad h_{\text{eff}} = \ln \frac{p_+}{p_-}. \quad (102)$$

The sum measures how readily a controlled opportunity produces any revision, whereas the ratio measures the direction favored conditional on revision.

Pooling all controlled updates gives

$$h_{\text{eff}} = 3.83, \quad \gamma_{\text{eff}} = 0.372,$$

with forward-to-reverse transition ratio $e^{h_{\text{eff}}} \simeq 46.2$. Thus motion toward the controller target is strongly favored among revisions, while most controlled opportunities still leave the focal vote unchanged. A stratified episode-level bootstrap gives the approximate 95% interval

$$h_{\text{eff}} \in [3.06, 4.88].$$

No further decomposition of h_{eff} is required by the finite-time thermodynamic theory. The same measured affinity enters the susceptibility, the controlled current, and the path identity $\Delta S_{\text{sys}} + hJ_c + I(n_k; Y_k) = \Sigma$.

Table 4: Empirical quantities connecting microscopic LLM dynamics to the single-affinity finite-compliance description.

Quantity	Estimate	Interpretation
h_{eff}	3.83	directional affinity
95% CI for h_{eff}	[3.06, 4.88]	bootstrap uncertainty
$e^{h_{\text{eff}}}$	46.2	forward/reverse transition ratio
γ_{eff}	0.372	probability scale for actual revision

D RESOURCE-CONSTRAINED CONTROL DESIGN

The theory in Section 5 separates three questions that are often conflated in controller design: how strongly the population can be moved, how coherently controller influence is converted into that motion, and how efficiently the finite-time sensing–actuation budget is converted into directed current. This appendix turns susceptibility, the two bounded efficiencies, and an explicit operational interaction cost into a practical design prescription.

D.1 MINIMUM-COST CONTROL AT PRESCRIBED RESPONSE

Let

$$\varphi = (q_c, b, \theta, \beta, \dots)$$

denote the tunable feedback-controller configuration, and let $\rho(n)$ be the population-state occupancy over the evaluation window. The microscopic channel parameters h and γ are treated separately: unless the implementation exposes a mechanism that changes them directly, they are calibrated properties of the population response rather than free design knobs.

Since the state-local susceptibility is defined conditionally on advocacy, the expected controller-induced response per round is

$$R(\varphi; h, \gamma) = \sum_{n=0}^N \rho(n) a_n(\varphi) \chi_{h,\gamma}\left(\frac{n}{N}\right). \quad (103)$$

A direct operational interaction cost is

$$C_{\text{op}}(\varphi) = q_c + b \sum_{n=0}^N \rho(n) a_n(\varphi), \quad (104)$$

where the first term counts the agents sensed every round and the second counts the expected number of controlled microscopic opportunities.

The most direct controller-design problem is therefore not to maximize an efficiency in isolation, but to achieve a prescribed behavioral response at minimum operational cost,

$$\varphi^* = \arg \min_{\varphi} C_{\text{op}}(\varphi) \quad \text{subject to} \quad R(\varphi; h, \gamma) \geq R_{\text{req}}. \quad (105)$$

Equivalently, for a fixed interaction budget one may maximize the achievable response,

$$\varphi^* = \arg \max_{\varphi} R(\varphi; h, \gamma) \quad \text{subject to} \quad C_{\text{op}}(\varphi) \leq C_{\text{max}}. \quad (106)$$

These formulations make the practical question explicit: once the required response has been fixed, how much sensing and actuation can be removed before performance falls below the desired level?

For the exact finite-compliance $q = 1$ reference,

$$\chi_{h,\gamma}(x) = [\sigma(h) - x] \Lambda_{b,\gamma}, \quad \Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b. \quad (107)$$

The marginal gain from one additional controlled opportunity is

$$\Lambda_{b+1,\gamma} - \Lambda_{b,\gamma} = \frac{\gamma}{N} \left(1 - \frac{\gamma}{N}\right)^b, \quad (108)$$

which decreases monotonically with b . Thus stronger actuation continues to increase the mean response toward the set point, but each additional controlled opportunity contributes less than the previous one. The same logic applies to sensing: increasing q_c is useful only insofar as the improved state estimate changes the policy in a way that improves the required response.

D.2 HOW THE TWO EFFICIENCIES GUIDE FEASIBLE DESIGNS

The operational problem above specifies how to purchase less control. The two bounded efficiencies from Section 5 then determine whether a feasible controller uses its influence and thermodynamic budget well.

The information–response efficiency is

$$\eta_{\text{IR}}(n) = \frac{2a_n(1 - a_n) \chi_{h,\gamma}(n/N)^2}{(\ln 2) T_\pi(n)} \leq 1, \quad (109)$$

with $T_\pi(n) = I(U_k; n_{k+1} \mid n_k = n)$. It measures whether the action-dependent change in the full next-state distribution is expressed as a coherent shift in mean target fraction. Among settings with comparable cost and response, one should prefer regimes in which η_{IR} remains high over the states actually visited.

The thermodynamic efficiency is

$$\eta_{\text{th},k} = \frac{hJ_{c,k}}{\Sigma_k - \Delta S_{\text{sys},k}} = \frac{hJ_{c,k}}{hJ_{c,k} + I(n_k; Y_k)} \leq 1. \quad (110)$$

Here $\Delta S_{\text{sys},k}$ is the transient storage term and $\Sigma_k - \Delta S_{\text{sys},k}$ is the non-storage finite-time control expenditure. The efficiency therefore reports the fraction of the generalized sensing–actuation expenditure expressed as affinity-weighted directed current. Its derivation from the forward/reverse path ratio is given in Appendix A.2.

Two practical rules follow directly. First, more sensing is not automatically better: if increasing q_c raises $I(n_k; Y_k)$ without a proportional increase in hJ_c , it lowers η_{th} . Second, more actuation is useful only while it continues to raise the required response; the exact susceptibility already shows diminishing marginal gains through $\Lambda_{b+1,\gamma} - \Lambda_{b,\gamma}$.

Taken together, the design prescription is

- (i) specify the minimum acceptable response or current,
 - (ii) minimize operational cost subject to that requirement,
 - (iii) use η_{IR} to reject incoherent action channels,
 - (iv) among feasible controllers, prefer larger η_{th} ,
 - (v) retain h and γ as separate directional and kinetic diagnostics.
- (111)

Thus susceptibility determines where control has leverage, the operational cost determines how much control must be purchased, η_{IR} determines whether controller influence becomes coherent mean motion, and η_{th} determines whether sensing and actuation are converted efficiently into directed current.

D.3 FINITE-TIME AND NONSTATIONARY INTERPRETATION

No stationarity assumption is required in the construction above. Let

$$\rho_k(n) = \Pr(n_k = n) \quad (112)$$

denote the transient population distribution at round k . Over a finite horizon of H rounds, the occupancy entering Eqs. (103)–(104) can be interpreted as the finite-time visitation distribution

$$\bar{\rho}_H(n) = \frac{1}{H} \sum_{k=0}^{H-1} \rho_k(n). \quad (113)$$

Hence the operational design problem averages over the transient states actually visited during the experiment rather than over an equilibrium distribution. Likewise, Eq. (94) retains the finite-time boundary contribution $\Delta S_{\text{sys},k}$, which accounts for the fact that the population ensemble changes during the feedback cycle. A stationary distribution is therefore only a special long-time limit of the present formulation, not an assumption required by it.